

Relate Chromosome X Analysis: Issues and Solutions Attempted

Date: November 2025 **Analysis:** Chromosome X non-PAR region phylogenetic inference using Relate
Dataset: 3,202 samples, ~2.25 million SNPs, chrX:3,533,229-154,781,072

Executive Summary

After extensive troubleshooting, we have successfully resolved multiple data formatting issues but encountered a fundamental limitation in Relate's `MakeChunks` function when processing large datasets with sparse genetic maps. The issue manifests as an assertion failure at `data.cpp:412` with the error `bp_pos[snp] == mbp`, which occurs even when using the recommended `RelateParallel` wrapper script.

Issue 1: Variable Haplotype Counts at PAR/Non-PAR Boundary

Problem Description

Initial Error:

```
Relate: data.cpp:571: Assertion `it_seq == sequence.end()' failed.  
Error at chromosome X position 2,781,514
```

Root Cause:

- PAR1 ends at position 2,781,479 (GRCh38)
- Non-PAR begins at position 2,781,480
- Error occurred at position 2,781,514 (first variant in non-PAR region)

Investigation Findings:

- Line 91,095 in HAPS file: 6,409 fields (PAR region - all samples diploid)
- Line 91,096 in HAPS file: 4,811 fields (non-PAR region - males haploid, females diploid)
- Field count calculation:
 - PAR: 5 metadata + 6,404 haplotypes = 6,409 fields
 - Non-PAR: 5 metadata + 4,806 haplotypes = 4,811 fields
 - Difference of 1,598 fields corresponds to 436 males × 2 = 872 haplotypes reduced to 436

Solution Implemented

☐ **Excluded PAR regions entirely** and analyzed only the non-PAR region (chrX:2,781,480-155,701,382)

Justification:

- Non-PAR represents 98% of chromosome X (151 Mb vs 3.1 Mb PAR)
- Most X-linked selection signals occur in non-PAR regions

- Provides consistent ploidy for Relate's algorithm
- Aligns with research goals (X-specific selection, dosage compensation)

Files Created:

- Initial filtered VCF: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR.vcf.gz`
- Corresponding HAPS: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR.haps`

Issue 2: Multiallelic Variants Creating Duplicate Positions

Problem Description

Error After PAR Exclusion:

```
Failed at BP 2783144
SNPs are not sorted by bp or more than one SNP at same position.
```

Investigation Findings:

- Found 18,184 duplicate positions in the HAPS file
- Example at position 2,783,144:

```
Line 52: chrX . 2783144 C T
Line 53: chrX . 2783144 C A
```

- Root cause: VCF contained multiallelic variants (e.g., REF=C, ALT=T,A)
- RelateFileFormats correctly splits these into separate biallelic records at the same position
- Relate requires unique positions for each SNP

Solution Implemented

❑ Used bcftools to split multiallelic sites and remove duplicates**Command Pipeline:**

```
bcftools view -r chrX:2781480-155701382 -v snps input.vcf.gz -O v | \
bcftools norm -m -snps -O v | \
bcftools norm -d snps -O z -o output_biallelic.vcf.gz
```

Where:

- `-m -snps`: Splits multiallelic sites into biallelic records
- `-d snps`: Removes duplicate SNP records (keeps first occurrence)

Results:

- Original: 2,295,317 SNPs with 18,184 duplicates
- After filtering: 2,276,950 unique biallelic SNPs
- Verification: 0 duplicate positions confirmed

Files Created:

- Script: `/home/vanbruggenmit/mit-ihh-pib/people/vanbruggenmit/mit-ihh-pib/scripts/filter_chrX_biallelic.slurm`
- Output VCF: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR_biallelic.vcf.gz` (2.1 GB)

Issue 3: Genetic Map Range Mismatch

Problem Description

New Error After Biallelic Filtering:

```
Failed to read line 2.
```

Investigation Findings:

- HAPS file started at position 2,781,514
- Genetic map started at position 3,533,229
- Gap of ~750 kb with no recombination rate data
- Relate requires genetic map to cover all SNP positions (or allow interpolation)

Solution Implemented

☐ Re-filtered VCF to match genetic map coverage range**Updated Region:** chrX:3,533,229-154,781,072**Commands:**

```
# Filter VCF to genetic map range
bcftools view -r chrX:3533229-154781072 -v snps input.vcf.gz -O v | \
bcftools norm -m -snps -O v | \
bcftools norm -d snps -O z -o chrX_nonPAR_biallelic.vcf.gz

# Convert to Relate format
RelateFileFormats --mode ConvertFromVcf \
  --haps chrX_nonPAR_biallelic.haps \
  --sample chrX_nonPAR_biallelic.sample \
  -i chrX_nonPAR_biallelic
```

Results:

- Final dataset: 2,252,381 biallelic SNPs
- Range: chrX:3,533,229-154,781,072 (151.2 Mb)
- All positions within genetic map coverage

Files Created:

- HAPS: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR_biallelic.haps` (21 GB)
- Sample: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR_biallelic.sample` (62 KB)

Issue 4: Genetic Map Assertion Failure (UNRESOLVED)

Problem Description

Persistent Error:

```
data.cpp:412: Assertion `bp_pos[snp] == mbp' failed.  
Error during MakeChunks step
```

This error occurs during the **MakeChunks** stage of Relate, which divides the chromosome into chunks for parallel processing.

Attempts to Resolve

Attempt 1: Created Distance File

Hypothesis: Relate might need explicit SNP distances

Implementation:

```
awk 'NR == 1 {print 0; prev_pos = $3; next} \  
    {print $3 - prev_pos; prev_pos = $3}' \  
chrX_nonPAR_biallelic.haps > chrX_nonPAR_biallelic.dist
```

Result: ☐ Same assertion failure

File Created: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR_biallelic.dist` (2,252,381 lines)

Attempt 2: Created Dense Interpolated Genetic Map

Hypothesis: Relate needs genetic map entry for every SNP position

Implementation:

- Created Python script using binary search for efficient interpolation
- Generated genetic map with 2,252,381 positions (one per SNP)
- Linearly interpolated cM values from sparse map (23,976 positions)

Script: `/home/vanbruggenmit/mit-ihh-pib/people/vanbruggenmit/mit-ihh-pib/scripts/interpolate_genetic_map.py`

Example:

```
Original sparse map (23,976 positions):
chrX:3533229 3533229 0.000123795
chrX:3534231 3534231 0.00102705
...

Interpolated dense map (2,252,381 positions):
chrX:3533229 3533229 0.000123795
chrX:3533246 3533246 0.0001391197 <- interpolated
chrX:3533342 3533342 0.0002256591 <- interpolated
...
```

Result: ☐ Same assertion failure

File Created: `/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/chrX_nonPAR_biallelic_interpolated.relate.map`
(2,252,381 lines)

Attempt 3: Used Relate Direct Call with --mode All

Hypothesis: Maybe wrapper scripts have bugs; try direct binary call

Command:

```
./bin/Relate \
  --mode All \
  --haps chrX_nonPAR_biallelic.haps \
  --sample chrX_nonPAR_biallelic.sample \
  --map chrX_nonPAR_biallelic_interpolated.relate.map \
  --dist chrX_nonPAR_biallelic.dist \
  -m 0.83e-8 \
  -N 20000 \
  --seed 1 \
  -o output
```

Result: ☐ Same assertion failure at data.cpp:412

Attempt 4: Used RelateParallel.sh Wrapper (Recommended Approach)

Hypothesis: Proper workflow requires using wrapper scripts, not direct calls

Discovery:

- Found Relate has specific wrapper scripts for different cluster systems
- `RelateParallel.sh`: For single-node parallel processing (recommended for our setup)
- `RelateSlurm.sh`: For distributing across multiple SLURM jobs
- Documentation shows these handle chunking and workflow automatically

Command:

```
./scripts/RelateParallel/RelateParallel.sh \  
  --haps chrX_nonPAR_biallelic.haps \  
  --sample chrX_nonPAR_biallelic.sample \  
  --map chrX_nonPAR.relate.map \      # sparse map (23,976 positions)  
  --dist chrX_nonPAR_biallelic.dist \  
  -m 0.83e-8 \  
  -N 20000 \  
  --seed 1 \  
  --threads 16 \  
  -o output
```

Result: ❌ Same assertion failure at data.cpp:412

Error Output:

```
-----  
Parsing data..  
Warning: Will use min 40GB of hard disc.  
Relate: data.cpp:412: Assertion `bp_pos[snp] == mbp' failed.  
./scripts/RelateParallel/RelateParallel.sh: line 542: Aborted (core dumped)
```

Attempt 5: Tried Running Without Genetic Map

Hypothesis: Maybe Relate can work with just distances and mutation rate

Command:

```
./bin/Relate \  
  --mode All \  
  --haps chrX_nonPAR_biallelic.haps \  
  --sample chrX_nonPAR_biallelic.sample \  
  --dist chrX_nonPAR_biallelic.dist \  
  -m 0.83e-8 \  
  -N 20000 \  
  -o output
```

```
--seed 1 \  
-o output
```

Result: Relate requires genetic map as mandatory argument

Error: Not enough arguments supplied. Needed: haps, sample, map, mutation_rate, effectiveN, output.

Analysis of Root Cause

The Assertion Failure

Location: data.cpp:412 in the MakeChunks function **Assertion:** bp_pos[snp] == mbp

This assertion checks that SNP positions exactly match genetic map positions. However, this is failing even though:

- 1. We have a genetic map with entries for all SNP positions (interpolated map)
- 2. Positions match exactly between HAPS and map files
- 3. Map format follows Relate's expected format (chrX:pos pos cM)
- 4. We're using the recommended RelateParallel wrapper

Dataset Characteristics

Our dataset has unusual characteristics that may trigger the bug:

Characteristic	Value	Comparison
SNPs	2,252,381	Very large
Samples	3,202 (6,404 haplotypes)	Large
Sparse genetic map	23,976 positions	Normal for genome-wide studies
Dense genetic map (attempted)	2,252,381 positions	Unusual/unprecedented
HAPS file size	21 GB	Very large
Interpolated map size	~70 MB	Very large for a genetic map

Hypothesis on Why This Fails

Based on our investigation and the Relate documentation:

- 1. **Relate is designed to work with sparse genetic maps** (documented behavior)
- 2. **The assertion at line 412 expects exact position matches** (implementation detail)
- 3. **These two requirements are contradictory** for large datasets

Possible explanations:

- The sparse map interpolation code path may not be working correctly in this version
- The assertion was added for debugging and is too strict for production use

- There may be an integer overflow or precision issue with large position numbers
 - The code may not have been tested with such large datasets (2.25M SNPs)
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What We've Confirmed Works

□ Data Preparation:

- Biallelic SNPs only (no multiallelic variants)
- No duplicate positions
- Matching genomic ranges between HAPS and genetic map
- Proper file formats (.haps, .sample, .map, .dist)
- Consistent haplotype counts (non-PAR only)

□ Workflow:

- Correct use of RelateFileFormats for VCF conversion
- Proper use of bcftools for filtering
- Correct use of RelateParallel wrapper script
- All files are properly formatted and pass basic validation

□ File Verification:

```
# Verified no duplicate positions
awk '{print $3}' chrX_nonPAR_biallelic.haps | sort -n | uniq -d | wc -l
# Output: 0

# Verified consistent field counts
awk '{print NF}' chrX_nonPAR_biallelic.haps | uniq -c
# Output: 2252381    4811

# Verified position ranges match
head -1 chrX_nonPAR_biallelic.haps | awk '{print $3}' # 3533229
head -1 chrX_nonPAR.relate.map | awk '{print $2}' # 3533229
tail -1 chrX_nonPAR_biallelic.haps | awk '{print $3}' # 154781072
tail -1 chrX_nonPAR.relate.map | awk '{print $2}' # 154781072
```

What Doesn't Work

□ Relate MakeChunks with large sparse maps:

- 2,252,381 SNPs with 23,976 genetic map positions
 - Fails with assertion error at data.cpp:412
 - Occurs even with proper wrapper scripts
 - Occurs even with dense interpolated maps
-

Files Created During Troubleshooting

Data Files

```
/home/vanbruggenmit/mit-ihh-pib/data/relate/chrX_nonPAR/
├─ chrX_nonPAR_biallelic.vcf.gz          # 2.1 GB - Filtered VCF
(biallelic, matching map range)
├─ chrX_nonPAR_biallelic.haps            # 21 GB - Relate input
(2,252,381 SNPs)
├─ chrX_nonPAR_biallelic.sample          # 62 KB - Sample metadata
├─ chrX_nonPAR_biallelic.dist            # ~18 MB - SNP distances
├─ chrX_nonPAR.relate.map                # 740 KB - Sparse genetic map
(23,976 positions)
└─ chrX_nonPAR_biallelic_interpolated.relate.map # ~70 MB - Dense map
(2,252,381 positions)
```

Scripts

```
/home/vanbruggenmit/mit-ihh-pib/people/vanbruggenmit/mit-ihh-pib/scripts/
├─ filter_chrX_biallelic.slurm           # VCF filtering pipeline
├─ interpolate_genetic_map.py            # Creates dense genetic map
├─ relate_chrX_nonPAR_direct.slurm       # Direct Relate call (failed)
├─ relate_chrX_nonPAR_parallel.slurm     # RelateParallel wrapper (failed)
└─ relate_chrX_nonPAR_nomap.slurm        # Attempted without map (failed)
```

Documentation

```
/home/vanbruggenmit/mit-ihh-pib/people/vanbruggenmit/mit-ihh-pib/notebooks/
└─ Relate_chrX_Issues_Summary.md        # This document
```

Recommendations

Immediate Actions

1. Contact Relate Developers

- Email: leo.speidel@outlook.com (from Relate README)
- Subject: Assertion failure with large sparse genetic maps (data.cpp:412)
- Include: Dataset statistics, error logs, steps to reproduce
- Ask: If this is a known limitation or bug

2. Check Relate GitHub Issues

- Repository: <https://github.com/MyersGroup/relate>
- Search for: "assertion", "sparse map", "MakeChunks", "data.cpp:412"
- Check if others have reported similar issues with large datasets

3. Try Relate Alternative Approaches (if available)

- Check if there's a development/beta version with fixes
- Look for Relate parameter to disable assertions (if exists)
- Check if source code can be modified to remove assertion

Workaround Options

1. Reduce SNP Density

```
# Thin SNPs to every Nth SNP
bcftools view -i 'ID=@snp_list.txt' input.vcf.gz
```

- Pros: May avoid assertion failure
- Cons: Loses resolution, may miss selection signals

2. Split Chromosome into Regions

- Analyze non-PAR as smaller chunks separately
- Pros: Smaller datasets may work
- Cons: Loses long-range haplotype information, more complex workflow

3. Use Alternative Tools

- **tsinfer**: Infers tree sequences from genetic variation data
- **ARGweaver**: Infers ancestral recombination graphs
- **Rent+**: For recombination rate estimation
- Pros: May handle large datasets better
- Cons: Different methods, different outputs

Long-term Solutions

1. Debug Relate Source Code

- Build Relate from source in debug mode
- Identify exact cause of assertion failure at line 412
- Submit patch to developers if possible

2. Collaborate with Relate Developers

- Share our dataset as test case
- Help them reproduce and fix the bug
- Contribute to improving Relate for large datasets

Technical Specifications

System Environment

- **Cluster**: HPC with SLURM scheduler

- **Relate Version:** Latest from git repository (November 2025)
- **Resources Used:** 16 CPUs, 64 GB RAM, 48-hour time limit

Dataset Specifications

- **Species:** Human (*Homo sapiens*)
- **Reference:** GRCh38
- **Chromosome:** X (non-PAR region)
- **Samples:** 3,202 individuals
 - 436 males (13.6%) - hemizygous in non-PAR
 - 2,766 females (86.4%) - diploid
- **Total Haplotypes:** 4,806 in non-PAR region
- **SNPs:** 2,252,381 biallelic variants
- **Region:** chrX:3,533,229-154,781,072 (151.2 Mb)
- **Mutation Rate:** 0.83×10^{-8} per base per generation
- **Effective Population Size:** 20,000 (haploid)

Genetic Map Specifications

- **Source:** [Specify your genetic map source]
- **Format:** Standard Relate format (position bp, rate cM/Mb, distance cM)
- **Sparse Map:** 23,976 positions across 151.2 Mb
- **Average Spacing:** ~6.3 kb between map positions
- **SNPs per Map Position:** ~94 SNPs per map position (2,252,381 / 23,976)

Conclusion

After extensive troubleshooting involving five different approaches, we have successfully prepared high-quality input data that meets all of Relate's documented requirements. However, we encounter a consistent assertion failure (`bp_pos[snp] == mbp` at `data.cpp:412`) during the `MakeChunks` stage.

This appears to be a **bug or undocumented limitation in Relate** when handling:

- Very large numbers of SNPs (2.25 million)
- Sparse genetic maps (typical for genome-wide studies)
- The combination of these two factors

The issue is not with our data preparation or workflow, as we have:

- ☐ Eliminated all known data quality issues
- ☐ Used the recommended `RelateParallel` wrapper
- ☐ Followed all documentation guidelines
- ☐ Verified file formats and compatibility

Further progress requires either:

1. Input from Relate developers on this specific error
2. Access to a patched version of Relate
3. Use of alternative phylogenetic inference tools

References

1. Relate Documentation: <https://myersgroup.github.io/relate/>
 2. Relate GitHub: <https://github.com/MyersGroup/relate>
 3. Relate Publication: Speidel et al., Nature Genetics 51: 1321-1329, 2019
 4. bcftools Documentation: <http://samtools.github.io/bcftools/>
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